Deriving the Energy-Momentum Tensor from the General Relativistic Action and compare it with its canonical counterpart in QFT

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Abstract

We explain how the energy-momentum tensor arises from the response of a relativistic matter action to variations of the spacetime metric. Starting from a curved-spacetime matter action, we derive the metric-variation definition of the Hilbert stress tensor, work out the key identities for varying the inverse metric and the invariant volume element $\sqrt{-g}$, and show explicitly how the tensor appears as the coefficient of the linear coupling $h_{\mu\nu}T^{\mu\nu}$ when the metric is expanded around flat spacetime. The note also compares this metric-derived tensor with the canonical Noether energy-momentum tensor obtained from translation invariance, explains why the canonical tensor need not be symmetric, and describes how improvement terms such as the Belinfante–Rosenfeld construction relate the two definitions in ordinary relativistic field theories. A scalar-field example is included to show the two approaches agreeing directly in a simple case.

From the field theory's perspective, we know that in general relativity the metric tensor $g_{\mu\nu}$ is a field, and that if we expand the general relativistic matter action around flat spacetime, then the terms linear in the metric perturbation $h_{\mu\nu}$ have the form

$$h_{\mu\nu}T^{\mu\nu}.$$

Part of the goal of our note is to carry out that calculation explicitly and show how the energy-momentum tensor is obtained from the variation of the action with respect to the metric tensor.

1. Matter action in curved spacetime

Let ϕ denote a generic matter field. It could be a scalar field, a spinor field, a gauge field, or a collection of fields. In general relativity, matter is coupled to a spacetime metric $g_{\mu\nu}$ through a matter action of the form

$$S_m[\phi, g] = \int d^4x\, \sqrt{-g}\, \mathcal{L}_m(\phi, \nabla\phi, g_{\mu\nu}),$$

where

$$g = \det(g_{\mu\nu}).$$

The factor $\sqrt{-g}$ is included so that the four-dimensional volume element

$$d^4x\,\sqrt{-g}$$

is invariant under coordinate transformations.

The central idea is that the metric tells matter how to measure distances, angles, and kinetic energies. Therefore the response of the matter action to a small change in the metric measures the local flow of energy and momentum. This response is precisely the energy-momentum tensor.

2. Definition from metric variation

The energy-momentum tensor is defined by varying the matter action with respect to the inverse metric $g^{\mu\nu}$:

$$\boxed{T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_m}{\delta g^{\mu\nu}}.}$$

Equivalently, the first-order variation of the matter action is

$$\boxed{\delta S_m = -\frac{1}{2}\int d^4x\,\sqrt{-g}\,T_{\mu\nu}\,\delta g^{\mu\nu}.}$$

One can also vary with respect to $g_{\mu\nu}$ instead. Since

$$g^{\mu\rho}g_{\rho\nu} = \delta^{\mu}{}_{\nu},$$

we vary both sides:

$$\delta g^{\mu\rho}g_{\rho\nu} + g^{\mu\rho}\delta g_{\rho\nu} = 0.$$

Multiplying by $g^{\nu\sigma}$ gives

$$\delta g^{\mu\sigma} = -g^{\mu\rho}g^{\sigma\nu}\delta g_{\rho\nu}.$$

Hence

$$\delta g^{\mu\nu} = -g^{\mu\rho}g^{\nu\sigma}\delta g_{\rho\sigma}.$$

Substituting this into the variation gives

$$\delta S_m = \frac{1}{2}\int d^4x\,\sqrt{-g}\,T^{\rho\sigma}\,\delta g_{\rho\sigma},$$

where

$$T^{\rho\sigma} = g^{\rho\mu}g^{\sigma\nu}T_{\mu\nu}.$$

Thus an equivalent definition is

$$\boxed{T^{\mu\nu} = \frac{2}{\sqrt{-g}}\frac{\delta S_m}{\delta g_{\mu\nu}}.}$$

3. Useful variation identities

To calculate $T_{\mu\nu}$ explicitly, we need the variation of $\sqrt{-g}$. A standard determinant identity says

$$\delta g = g\, g^{\mu\nu} \delta g_{\mu\nu}.$$

Therefore

$$\delta\sqrt{-g} = \frac{1}{2\sqrt{-g}}\delta(-g) = -\frac{1}{2\sqrt{-g}}\delta g.$$

Using $\delta g = g\, g^{\mu\nu}\delta g_{\mu\nu}$, we get

$$\delta\sqrt{-g} = -\frac{1}{2\sqrt{-g}} g\, g^{\mu\nu}\delta g_{\mu\nu}.$$

Since $g = -(\sqrt{-g})^2$, this becomes

$$\boxed{\delta\sqrt{-g} = \frac{1}{2}\sqrt{-g}\, g^{\mu\nu}\delta g_{\mu\nu}.}$$

If instead we vary the inverse metric, then

$$g^{\mu\nu}\delta g_{\mu\nu} = -g_{\mu\nu}\delta g^{\mu\nu},$$

so

$$\boxed{\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\, g_{\mu\nu}\delta g^{\mu\nu}.}$$

4. General formula when the Lagrangian contains no metric derivatives

Suppose the matter Lagrangian depends on the metric algebraically, but not on derivatives of the metric:

$$\mathcal{L}_m = \mathcal{L}_m(\phi, \partial\phi, g^{\mu\nu}).$$

Then

$$S_m = \int d^4x\, \sqrt{-g}\, \mathcal{L}_m.$$

Varying with respect to $g^{\mu\nu}$ gives

$$\delta S_m = \int d^4x\, \left[\delta\sqrt{-g}\,\mathcal{L}_m + \sqrt{-g}\,\frac{\partial\mathcal{L}_m}{\partial g^{\mu\nu}}\delta g^{\mu\nu}\right].$$

Using

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\, g_{\mu\nu}\delta g^{\mu\nu},$$

we obtain

$$\delta S_m = \int d^4x\, \sqrt{-g}\, \left[-\frac{1}{2}g_{\mu\nu}\mathcal{L}_m + \frac{\partial\mathcal{L}_m}{\partial g^{\mu\nu}}\right]\delta g^{\mu\nu}.$$

Compare this with the defining relation

$$\delta S_m = -\frac{1}{2}\int d^4x\,\sqrt{-g}\,T_{\mu\nu}\delta g^{\mu\nu}.$$

Therefore

$$-\frac{1}{2}T_{\mu\nu} = -\frac{1}{2}g_{\mu\nu}\mathcal{L}_m + \frac{\partial\mathcal{L}_m}{\partial g^{\mu\nu}}.$$

Multiplying by -2 gives the useful formula

$$\boxed{T_{\mu\nu} = g_{\mu\nu}\mathcal{L}_m - 2\frac{\partial\mathcal{L}_m}{\partial g^{\mu\nu}}.}$$

This is the stress-energy tensor obtained from the general relativistic matter action. It is automatically symmetric because it is obtained by varying with respect to the symmetric tensor $g^{\mu\nu}$.

5. Expansion around flat spacetime

Now let's expand the metric around the Minkowski metric:

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu},$$

where κ is a coupling constant. In many general relativity conventions,

$$\kappa = \sqrt{32\pi G}.$$

And we can use a convention of the schematic form

$$g_{\mu\nu} = \eta_{\mu\nu} + \sqrt{G_N}\,h_{\mu\nu}.$$

The precise numerical coefficient is conventional; the important point is that $h_{\mu\nu}$ is the small metric perturbation.

Since

$$\delta g_{\mu\nu} = \kappa\,\delta h_{\mu\nu},$$

the first-order change in the matter action is

$$\delta S_m = \frac{1}{2}\int d^4x\,\sqrt{-g}\,T^{\mu\nu}\delta g_{\mu\nu} = \frac{\kappa}{2}\int d^4x\,\sqrt{-g}\,T^{\mu\nu}\delta h_{\mu\nu}.$$

Evaluating this to leading order around flat spacetime gives

$$\sqrt{-g} = 1 + O(h), \qquad T^{\mu\nu} = T^{\mu\nu}_{\text{flat}} + O(h).$$

Therefore the part of the action linear in $h_{\mu\nu}$ is

$$\boxed{S_m[\phi, \eta + \kappa h] = S_m[\phi, \eta] + \frac{\kappa}{2}\int d^4x\,h_{\mu\nu}T^{\mu\nu}_{\text{flat}} + O(h^2).}$$

Thus after expanding the general relativistic action around flat spacetime, the linear coupling of the metric perturbation to matter has the form

$$h_{\mu\nu}T^{\mu\nu}.$$

Thus $T^{\mu\nu}$ is the object that couples directly to the gravitational field $h_{\mu\nu}$.

6. Explicit example: real scalar field

Consider a real scalar field ϕ with curved-spacetime matter action

$$S_m = \int d^4x \sqrt{-g}\,\mathcal{L}_m,$$

where

$$\mathcal{L}_m = -\frac{1}{2} g^{\rho\sigma}\partial_\rho\phi\partial_\sigma\phi - V(\phi).$$

This sign choice is consistent with the convention

$$\delta S_m = -\frac{1}{2}\int d^4x \sqrt{-g}\, T_{\mu\nu}\delta g^{\mu\nu}.$$

We compute

$$T_{\mu\nu} = g_{\mu\nu}\mathcal{L}_m - 2\frac{\partial\mathcal{L}_m}{\partial g^{\mu\nu}}.$$

Only the kinetic term depends on $g^{\mu\nu}$. Since $g^{\rho\sigma}$ is symmetric, its variation satisfies

$$\delta g^{\rho\sigma}\partial_\rho\phi\partial_\sigma\phi = \delta g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi.$$

Therefore

$$\frac{\partial\mathcal{L}_m}{\partial g^{\mu\nu}} = -\frac{1}{2}\partial_\mu\phi\partial_\nu\phi.$$

Substituting into the formula gives

$$\boxed{T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - g_{\mu\nu}\left[\frac{1}{2}g^{\rho\sigma}\partial_\rho\phi\partial_\sigma\phi + V(\phi)\right].}$$

Equivalently, since

$$g^{\rho\sigma}\partial_\rho\phi\partial_\sigma\phi = (\nabla\phi)^2,$$

we can write

$$T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - g_{\mu\nu}\left[\frac{1}{2}(\nabla\phi)^2 + V(\phi)\right].$$

Different metric-signature conventions move some signs between the scalar Lagrangian and the definition of $T_{\mu\nu}$, but the calculation is always the same: vary the matter action with respect to the metric and read off the tensor that multiplies the metric variation.

7. Conservation from diffeomorphism invariance

The energy-momentum tensor obtained this way is conserved because the matter action is invariant under diffeomorphisms. Infinitesimally, a coordinate shift generated by a vector field ξ^μ changes the metric by the Lie derivative

$$\delta g_{\mu\nu} = \nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu.$$

Using

$$\delta S_m = \frac{1}{2}\int d^4x\,\sqrt{-g}\,T^{\mu\nu}\delta g_{\mu\nu},$$

we get

$$\begin{aligned}\delta S_m &= \frac{1}{2}\int d^4x\,\sqrt{-g}\,T^{\mu\nu}\left(\nabla_\mu\xi_\nu + \nabla_\nu\xi_\mu\right)\\ &= \int d^4x\,\sqrt{-g}\,T^{\mu\nu}\nabla_\mu\xi_\nu,\end{aligned}$$

where we used the symmetry $T^{\mu\nu} = T^{\nu\mu}$. Integrating by parts,

$$\int d^4x\,\sqrt{-g}\,T^{\mu\nu}\nabla_\mu\xi_\nu = -\int d^4x\,\sqrt{-g}\,\left(\nabla_\mu T^{\mu\nu}\right)\xi_\nu,$$

up to a boundary term. Diffeomorphism invariance means $\delta S_m = 0$ for arbitrary ξ_ν, so

$$\boxed{\nabla_\mu T^{\mu\nu} = 0.}$$

In flat spacetime, where ∇_μ reduces to ∂_μ, this becomes

$$\boxed{\partial_\mu T^{\mu\nu} = 0.}$$

This is the local conservation law for energy and momentum.

8. Summary on the GR part

The full calculation can be summarized as follows:

$$S_m[g,\phi] = \int d^4x\,\sqrt{-g}\,\mathcal{L}_m,$$

$$\delta S_m = -\frac{1}{2}\int d^4x\,\sqrt{-g}\,T_{\mu\nu}\delta g^{\mu\nu},$$

so

$$\boxed{T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_m}{\delta g^{\mu\nu}}.}$$

Equivalently,

$$\boxed{T^{\mu\nu} = \frac{2}{\sqrt{-g}}\frac{\delta S_m}{\delta g_{\mu\nu}}.}$$

Expanding around flat spacetime,

$$g_{\mu\nu} = \eta_{\mu\nu} + \kappa h_{\mu\nu},$$

gives

$$\boxed{S_m[\eta+\kappa h,\phi] = S_m[\eta,\phi] + \frac{\kappa}{2}\int d^4x\,h_{\mu\nu}T^{\mu\nu} + O(h^2).}$$

Therefore the energy-momentum tensor is exactly the coefficient of the linear coupling between matter and the metric perturbation.

9. Canonical energy-momentum tensor from translations

The image derives the energy-momentum tensor in ordinary classical field theory using Noether's theorem. Let us reproduce that derivation carefully and then compare it with the metric-variation definition used above.

Work first in flat spacetime, with coordinates x^μ and fields $\phi_n(x)$. The index n labels all fields in the theory. Suppose the Lagrangian density is

$$\mathcal{L} = \mathcal{L}(\phi_n, \partial_\mu \phi_n).$$

Under a constant spacetime translation

$$x^\mu \longrightarrow x'^\mu = x^\mu + \epsilon^\mu,$$

the field changes, at the same coordinate point, by

$$\delta\phi_n(x) = \epsilon^\nu \partial_\nu \phi_n(x).$$

Here the sign depends on whether one uses an active or passive convention for the translation. The final conserved tensor may therefore differ by an overall sign or by an index-order convention. We follow the convention shown in the image.

Because $\mathcal{L}$ is a scalar, it transforms similarly:

$$\delta\mathcal{L} = \epsilon^\nu \partial_\nu \mathcal{L}.$$

On the other hand, using the equations of motion, the variation of the Lagrangian density can be written as a total derivative. Indeed,

$$\begin{aligned}
\delta\mathcal{L} &= \sum_n \frac{\partial \mathcal{L}}{\partial \phi_n}\delta\phi_n + \sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)}\delta(\partial_\mu \phi_n) \\
&= \sum_n \frac{\partial \mathcal{L}}{\partial \phi_n}\delta\phi_n + \sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)}\partial_\mu \delta\phi_n.
\end{aligned}$$

Using the Euler–Lagrange equations,

$$\frac{\partial \mathcal{L}}{\partial \phi_n} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \right),$$

we get

$$\begin{aligned}
\delta\mathcal{L} &= \sum_n \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \right)\delta\phi_n + \sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)}\partial_\mu \delta\phi_n \\
&= \partial_\mu \left[\sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)}\delta\phi_n \right].
\end{aligned}$$

Now substitute the translation variation

$$\delta\phi_n = \epsilon^\nu \partial_\nu \phi_n.$$

Then

$$\delta\mathcal{L} = \epsilon^\nu \partial_\mu \left[\sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n \right].$$

But we also have

$$\delta\mathcal{L} = \epsilon^\nu \partial_\nu \mathcal{L} = \epsilon^\nu \partial_\mu (\delta^\mu{}_\nu \mathcal{L}).$$

Since ϵ^ν is arbitrary, we obtain

$$\partial_\mu \left[\sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - \delta^\mu{}_\nu \mathcal{L} \right] = 0.$$

Thus the canonical Noether energy-momentum tensor is

$$\boxed{T^\mu{}_{\nu,\text{can}} = \sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - \delta^\mu{}_\nu \mathcal{L}.}$$

Lowering the first index with the flat metric gives the convention used in the image:

$$\boxed{T^{\text{can}}_{\mu\nu} = \sum_n \frac{\partial \mathcal{L}}{\partial(\partial^\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu} \mathcal{L}.}$$

The conservation law is

$$\boxed{\partial_\mu T^\mu{}_{\nu,\text{can}} = 0.}$$

This conservation law follows from invariance under spacetime translations. There are four translations, one for each value of ν, and therefore four Noether currents $T^\mu{}_{\nu,\text{can}}$.

10. Relation to the metric energy-momentum tensor

The energy-momentum tensor derived earlier from the general relativistic matter action was

$$\boxed{T^{\text{metric}}_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}.}$$

This is often called the Hilbert energy-momentum tensor or the metric energy-momentum tensor.

The two derivations answer two slightly different questions:

method	symmetry/source	object obtained
Noether translation method	flat-spacetime translations	$T^\mu{}_{\nu,\text{can}}$
metric variation method	response to $g_{\mu\nu}$	$T^{\text{metric}}_{\mu\nu}$

The canonical tensor is the Noether current for translations. It is the quantity whose conservation expresses energy and momentum conservation in flat spacetime:

$$\partial_\mu T^\mu{}_{\nu,\text{can}} = 0.$$

The metric energy-momentum tensor is the source of gravity. It is the quantity that appears when matter is coupled to a weak gravitational field:

$$S_m[\eta + \kappa h, \phi] = S_m[\eta, \phi] + \frac{\kappa}{2}\int d^4x\, h_{\mu\nu} T^{\mu\nu}_{\text{metric}} + O(h^2).$$

Therefore the metric energy-momentum tensor answers the question: what does the gravitational field couple to?

The important relation is that, for ordinary relativistic field theories, the metric energy-momentum tensor agrees with the canonical tensor up to improvement terms:

$$\boxed{T^{\mu\nu}_{\text{metric}} = T^{\mu\nu}_{\text{can}} + \partial_\rho B^{\rho\mu\nu},}$$

where $B^{\rho\mu\nu}$ is antisymmetric in its first two indices,

$$B^{\rho\mu\nu} = -B^{\mu\rho\nu}.$$

Because of this antisymmetry,

$$\partial_\mu \partial_\rho B^{\rho\mu\nu} = 0,$$

so the improvement term is identically conserved. Hence

$$\partial_\mu T^{\mu\nu}_{\text{can}} = 0 \qquad \Longrightarrow \qquad \partial_\mu T^{\mu\nu}_{\text{metric}} = 0,$$

assuming the equations of motion hold.

The improvement term also does not change the total conserved four-momentum when fields fall off sufficiently fast at spatial infinity. The total momentum is

$$P^\nu = \int d^3x\, T^{0\nu}.$$

If

$$T^{0\nu}_{\text{metric}} = T^{0\nu}_{\text{can}} + \partial_i B^{i0\nu},$$

then

$$P^\nu_{\text{metric}} - P^\nu_{\text{can}} = \int d^3x\, \partial_i B^{i0\nu} = \int_{\partial\mathbb{R}^3} dS_i\, B^{i0\nu}.$$

For fields that vanish at infinity, the boundary integral vanishes. Therefore

$$\boxed{P^\nu_{\text{metric}} = P^\nu_{\text{can}}.}$$

Thus the canonical and metric energy-momentum tensors may differ locally, but they give the same conserved total energy and momentum under standard boundary conditions.

11. Why the canonical tensor is not always symmetric

The canonical tensor is obtained from translations only. Translation invariance imposes conservation,

$$\partial_\mu T^\mu{}_{\nu,\text{can}} = 0,$$

but it does not require symmetry in the two spacetime indices. Thus in general

$$T^{\mu\nu}_{\text{can}} \neq T^{\nu\mu}_{\text{can}}.$$

This is why the image says that the energy-momentum tensor defined by Noether's translation argument is not necessarily symmetric.

By contrast, the metric tensor is symmetric automatically, because it is defined by varying with respect to the symmetric metric $g^{\mu\nu}$:

$$g^{\mu\nu} = g^{\nu\mu}.$$

Therefore

$$\boxed{T^{\text{metric}}_{\mu\nu} = T^{\text{metric}}_{\nu\mu}.}$$

For theories with spin, the antisymmetric part of the canonical tensor is related to the spin current. Lorentz invariance implies that one can add a total-derivative improvement term to construct a symmetric tensor. This is the Belinfante–Rosenfeld tensor:

$$\boxed{T^{\mu\nu}_{\text{Bel}} = T^{\mu\nu}_{\text{can}} + \partial_\rho B^{\rho\mu\nu}.}$$

For standard minimally coupled theories, this improved tensor agrees with the metric energy-momentum tensor after going back to flat spacetime:

$$\boxed{T^{\mu\nu}_{\text{Bel}} = T^{\mu\nu}_{\text{metric}}.}$$

This is the precise sense in which the canonical and general relativistic definitions are equivalent: not necessarily as raw local formulas, but after the canonical tensor is improved by allowed total-derivative terms.

12. Explicit comparison for a scalar field

For a real scalar field in flat spacetime, take

$$\mathcal{L} = \frac{1}{2}\eta^{\rho\sigma}\partial_\rho\phi\partial_\sigma\phi - V(\phi)$$

with the mostly-minus metric. The canonical tensor is

$$T^{\mu}{}_{\nu,\text{can}} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\partial_\nu\phi - \delta^{\mu}{}_{\nu}\mathcal{L}.$$

Since

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = \partial^\mu\phi,$$

we find

$$T^{\mu}{}_{\nu,\text{can}} = \partial^\mu\phi\partial_\nu\phi - \delta^{\mu}{}_{\nu}\left[\frac{1}{2}\partial_\rho\phi\partial^\rho\phi - V(\phi)\right].$$

Raising the second index gives

$$\boxed{T^{\mu\nu}_{\text{can}} = \partial^\mu\phi\partial^\nu\phi - \eta^{\mu\nu}\left[\frac{1}{2}\partial_\rho\phi\partial^\rho\phi - V(\phi)\right].}$$

This tensor is already symmetric because

$$\partial^\mu\phi\partial^\nu\phi = \partial^\nu\phi\partial^\mu\phi.$$

Now compute the metric tensor. Couple the scalar field to a curved metric:

$$S_m = \int d^4x\,\sqrt{-g}\left[\frac{1}{2}g^{\rho\sigma}\partial_\rho\phi\partial_\sigma\phi - V(\phi)\right].$$

Using the metric-variation convention appropriate to this sign choice,

$$\delta S_m = \frac{1}{2}\int d^4x\,\sqrt{-g}\,T_{\mu\nu}\delta g^{\mu\nu},$$

one obtains

$$\boxed{T^{\text{metric}}_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - g_{\mu\nu}\left[\frac{1}{2}g^{\rho\sigma}\partial_\rho\phi\partial_\sigma\phi - V(\phi)\right].}$$

Returning to flat spacetime gives

$$\boxed{T^{\mu\nu}_{\text{metric}} = T^{\mu\nu}_{\text{can}}}$$

for the minimally coupled scalar field.

So for scalar fields the two methods give the same local tensor directly. For fields with spin, gauge fields, or theories where the Lagrangian contains total derivative ambiguities, the raw canonical tensor may differ from the metric tensor by improvement terms. The improved canonical tensor is the one that matches the GR metric variation result.

13. Final conceptual picture

The canonical derivation starts with flat-spacetime translation invariance:

$$\text{translations} \quad \Longrightarrow \quad \partial_\mu T^\mu{}_{\nu,\text{can}} = 0.$$

It identifies energy and momentum as Noether charges.

The general relativistic derivation starts by allowing the metric to become a dynamical background field:

$$\text{vary } g_{\mu\nu} \quad \Longrightarrow \quad T^{\text{metric}}_{\mu\nu} = \text{source coupled to gravity}.$$

It identifies energy and momentum as the source of spacetime curvature.

The relationship is therefore

$$\boxed{\text{canonical tensor} \xrightarrow{\text{symmetrize/improve}} \text{metric tensor} \xrightarrow{\text{couples to } h_{\mu\nu}} \text{source for gravity.}}$$

In simple scalar theories, no improvement is needed. In more general theories, the canonical tensor is conserved but not necessarily symmetric, while the metric energy-momentum tensor is symmetric and is the object that appears in general relativity.